

Basic Principles of Medicine 2

Module : Digestion and Metabolism

DLA Title: Structure and function - Carbohydrates

Dr. Sharmila Upadhyia
Email: shupadhyia@sgu.edu
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Assigned Reading

- Lippincott's Biochemistry 9th Edition
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
 - Lippincott's 8th Edition Chap 7 [Introduction to Carbohydrates | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#) (Concept map Fig. 7.12)
 - **Lippincott's Biochemistry 9th Edition: Questions 7.1-7.4**
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- Review of organic chemistry
 - Review chemistry to understand enzyme nomenclature in carbohydrate digestion and metabolism
 - **Focus on medical and nutritional significance of carbohydrates**

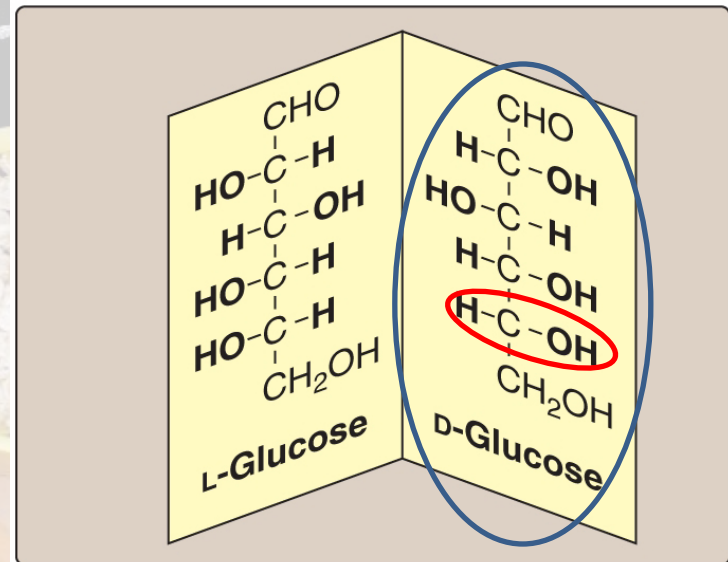
Objectives: DLA: Structure and function - Carbohydrates

- After DM Lecture 01

SOM. MK.I. BPM2.3. DM.2.BCHM.1099	Outline classification of monosaccharides based on number of carbon atoms and functional group (aldoses and ketoses)
SOM. MK.I. BPM2.3. DM.2.BCHM.1100	Differentiate between: Reducing and non-reducing sugars
SOM. MK.I. BPM2.3. DM.2.BCHM.1101	Define epimer using galactose as an example
SOM. MK.I. BPM2.3. DM.2.BCHM.1102	Compare and contrast the bonds present in: Monosaccharides (glucose, galactose, fructose); Disaccharides (maltose, sucrose, lactose); Polysaccharides (starch, glycogen and cellulose)
SOM. MK.I. BPM2.3. DM.2.BCHM.1103	Identify significance of sugar alcohols (sorbitol, galactitol) and sugar acids (glucuronic acid)
SOM. MK.I. BPM2.3. DM.2.BCHM.1104	Compare sucrose with High Fructose Corn Syrup (HFCS)
SOM. MK.I. BPM2.3. DM.2.BCHM.1105	Review clinical significance of detecting reducing sugars in urine in diabetes mellitus and inherited disorders of fructose and galactose metabolism

Overview of functions of carbohydrates

- Energy in diet (45-65%)
- Storage form: Glycogen in liver and muscle
- Dietary fiber: Cellulose
- Glucose in blood (Brain; RBCs, WBCs, tissues of eye)
- Connective tissue/ cell membrane
- D-forms of sugars



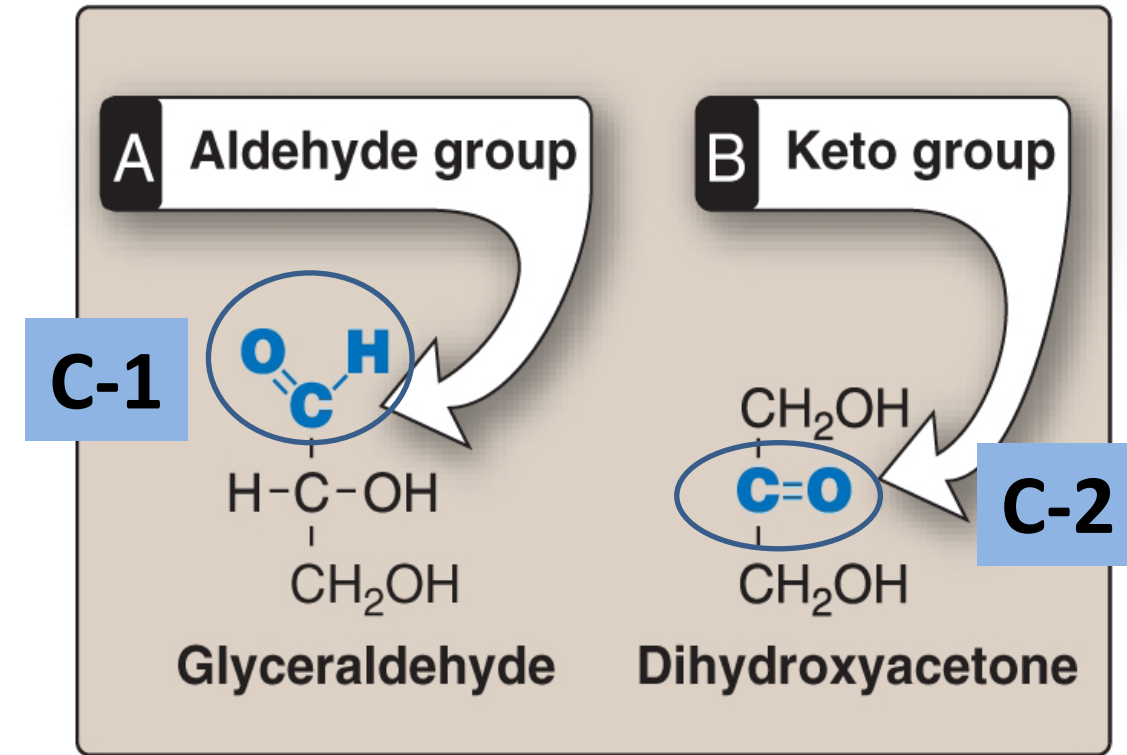
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Monosaccharides: Functional group and C-atoms

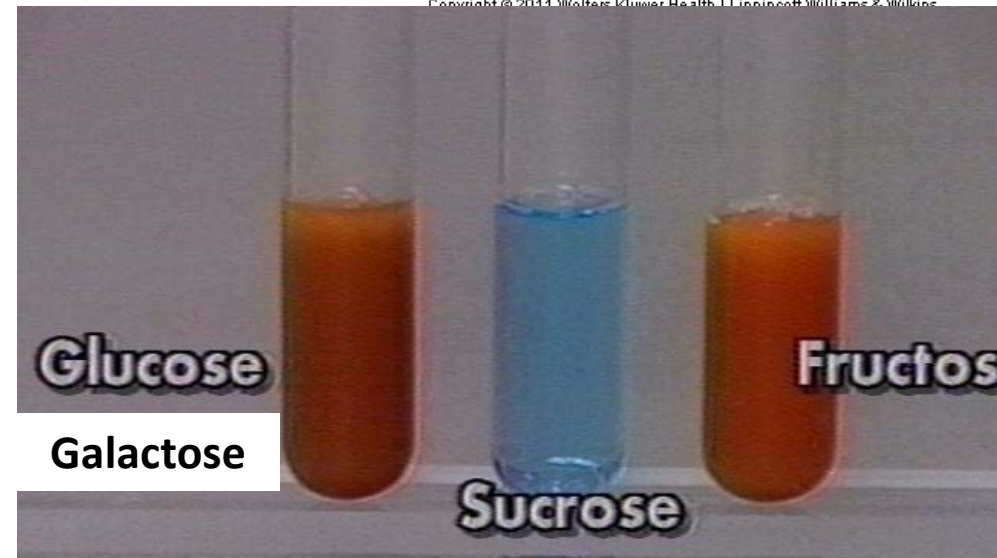
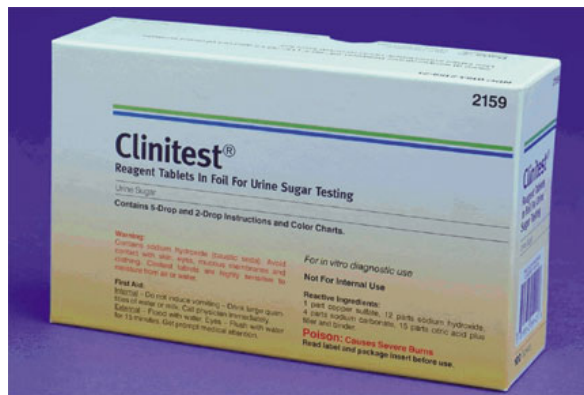
	Aldose (C-1 Aldehyde)	Ketose (C-2 Keto)
3 carbon (Triose)	Glyceraldehyde	Dihydroxyacetone
5 carbon (Pentose)	Ribose Xylose	Ribulose Xylulose
6 carbon (Hexose)	Glucose Galactose Mannose	Fructose

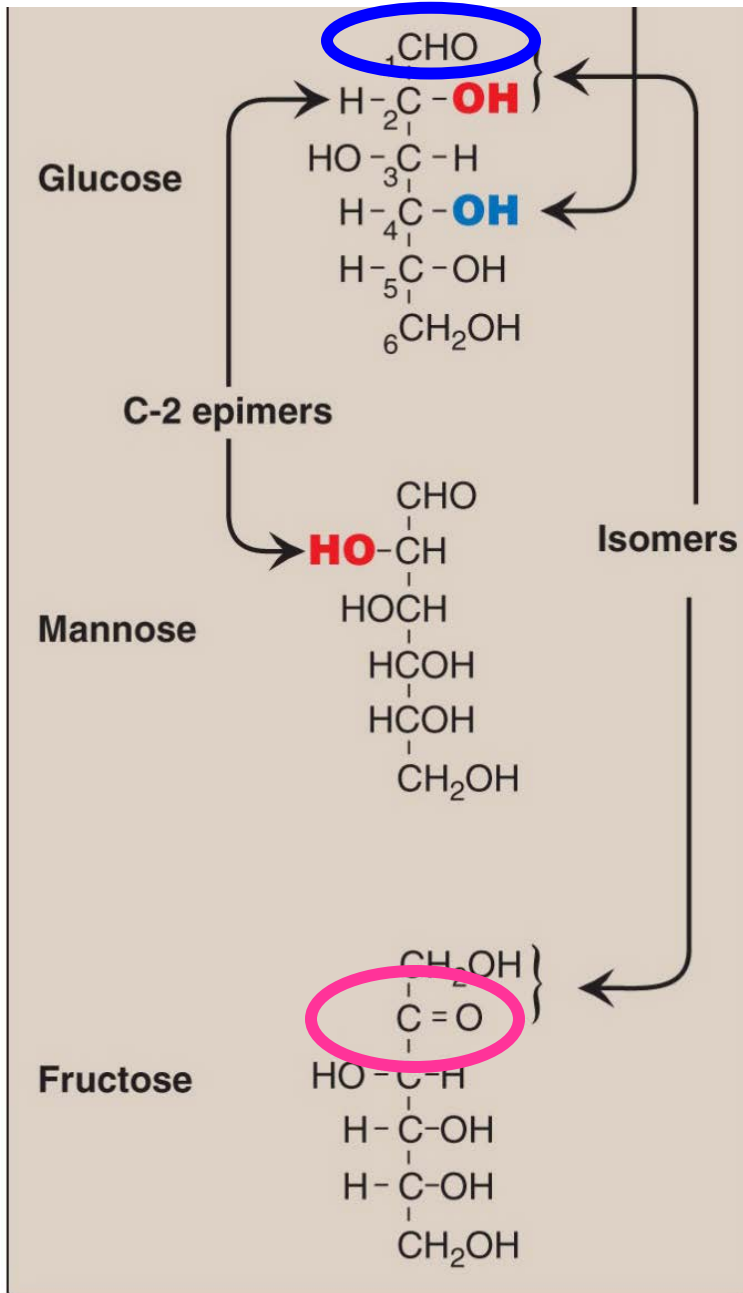
Reducing sugars

- Free aldehyde or keto group
- **Reducing sugars** in urine
 - Convert cupric ions to cuprous ions
- Aldoses vs Ketoses



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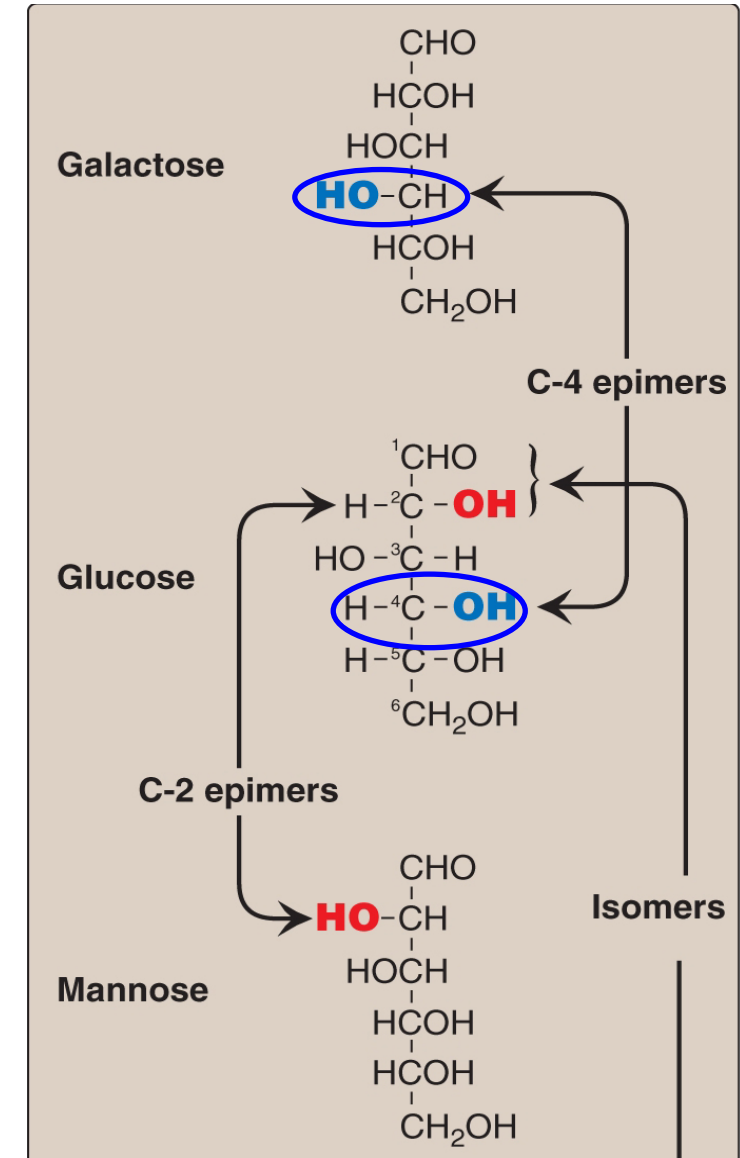
Glucose vs Fructose

(Aldohexose vs Ketohexose)

- Phosphohexose isomerase (glycolysis)
- **Reducing sugars**

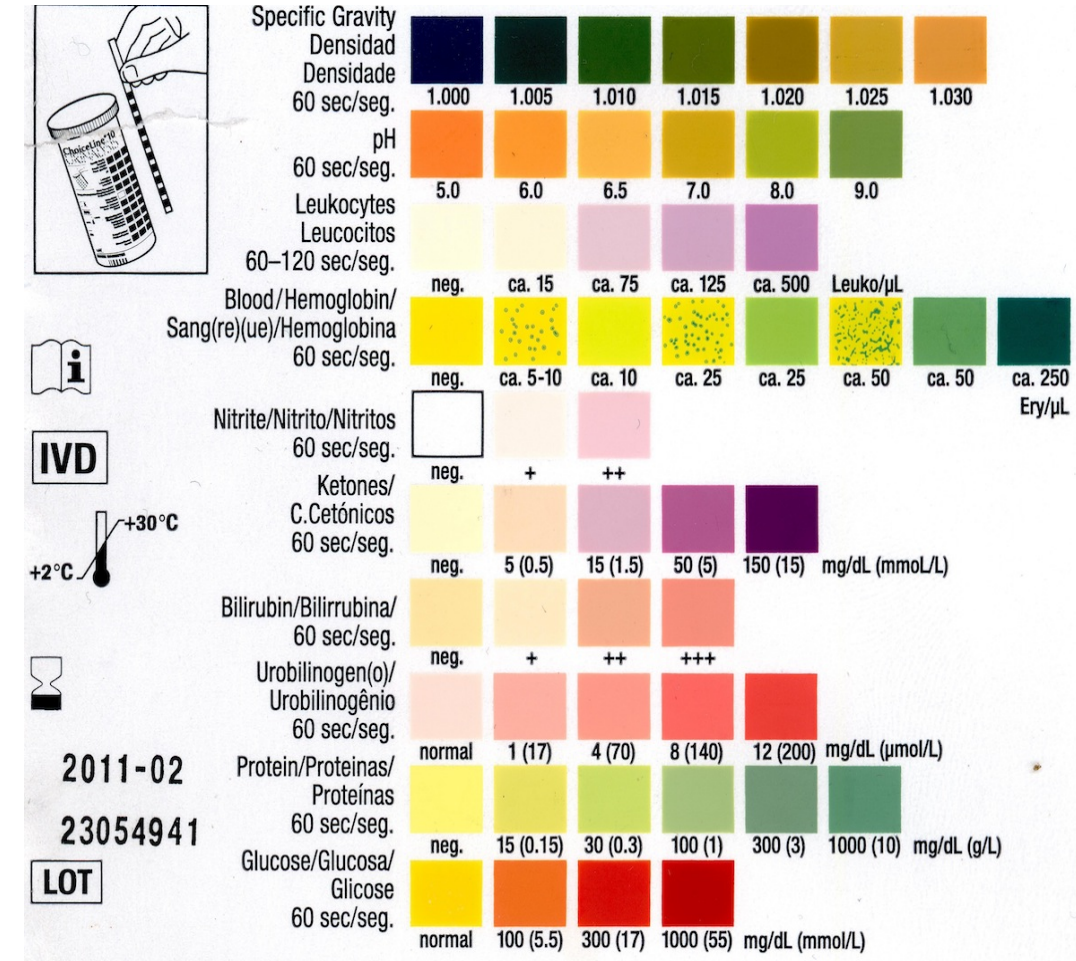
Galactose

- **C-4** epimer of glucose
- Aldohexose
- Epimerases (galactose metabolism)
- Avoid galactose: Galactosemia



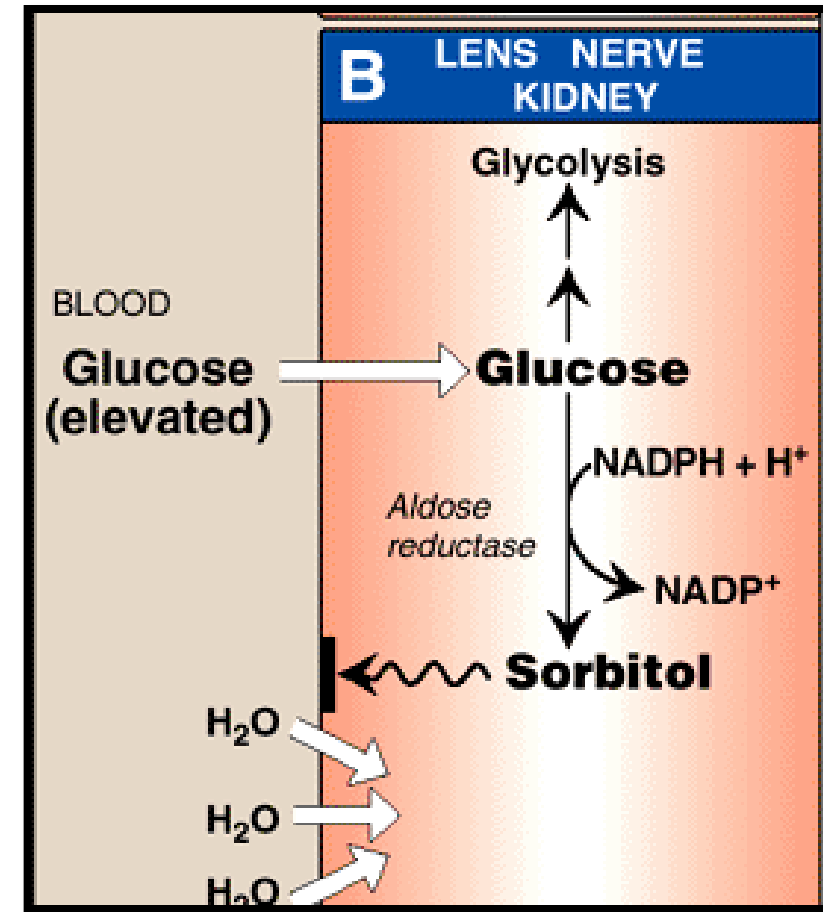
Sugars in urine: Clinical significance

- Sugars **absent** in urine of a normal person
- Reducing property
 - Benedicts test or Clinitest
- Sugars in urine:
 - **Glucose: Diabetes mellitus**
 - **Fructose**
 - **Galactose**
 - Lactose intolerance (no sugar in urine)



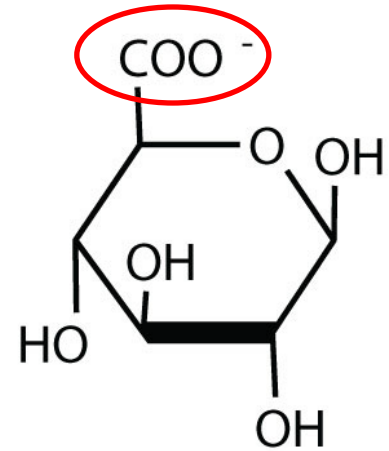
Polyols (sugar alcohols)

- **Sorbitol**: Sugar alcohol from glucose
 - **Diabetes mellitus**
- **Galactitol** formed from galactose
 - **Galactosemia**
- **Inositol** present in IP3 (second messenger)



Glucuronic acid

- Oxidation of C-6 of glucose
- Negatively charged
- UDP-glucuronic acid: Conjugation reactions (bilirubin)



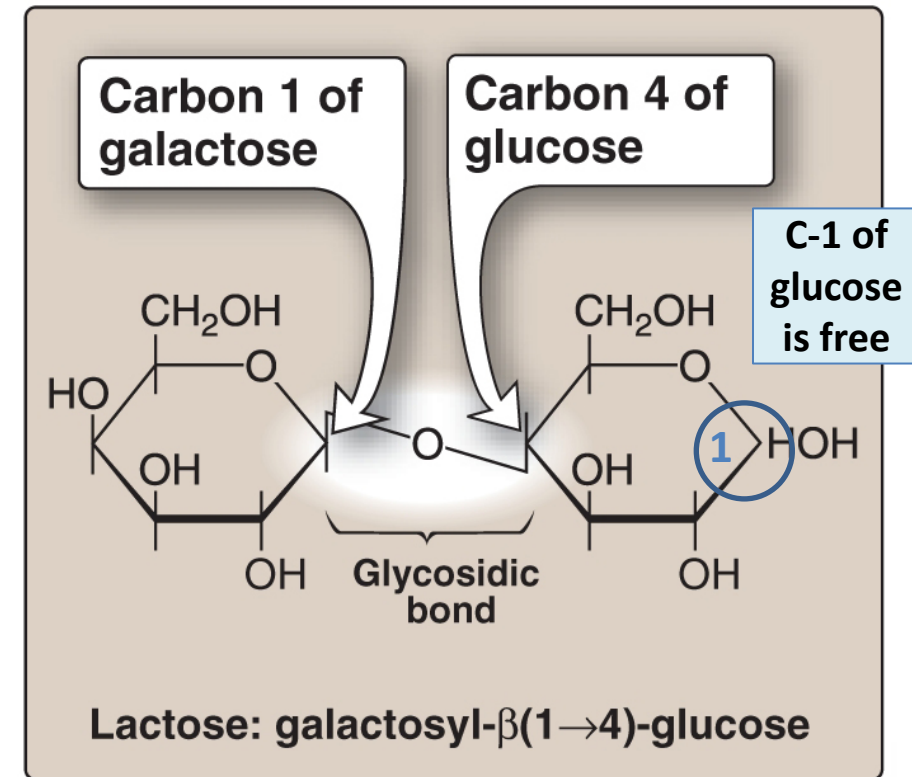
Glucuronic acid

Classification of carbohydrates

- **Monosaccharides: Monomers**
- Based on number of monomers:
 - Monosaccharides
 - **Disaccharides**
 - Oligosaccharides
 - Polysaccharides

Lactose

- Galactose + Glucose
- $\beta 1 \rightarrow 4$ glycosidic linkage
- Reducing sugar
- Intestinal **lactase** (brush border disaccharidase)
- **NOT found in urine in lactose intolerance**
- Avoid Lactose: Lactose intolerance and galactosemia

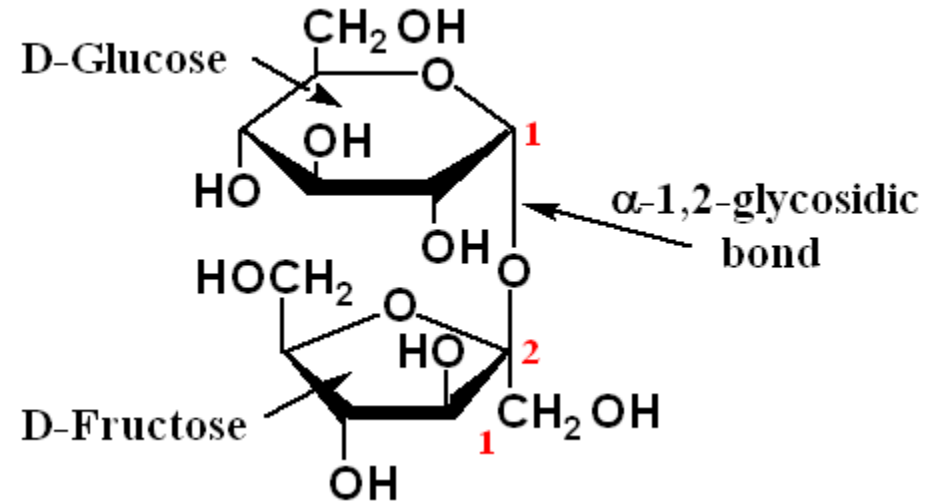


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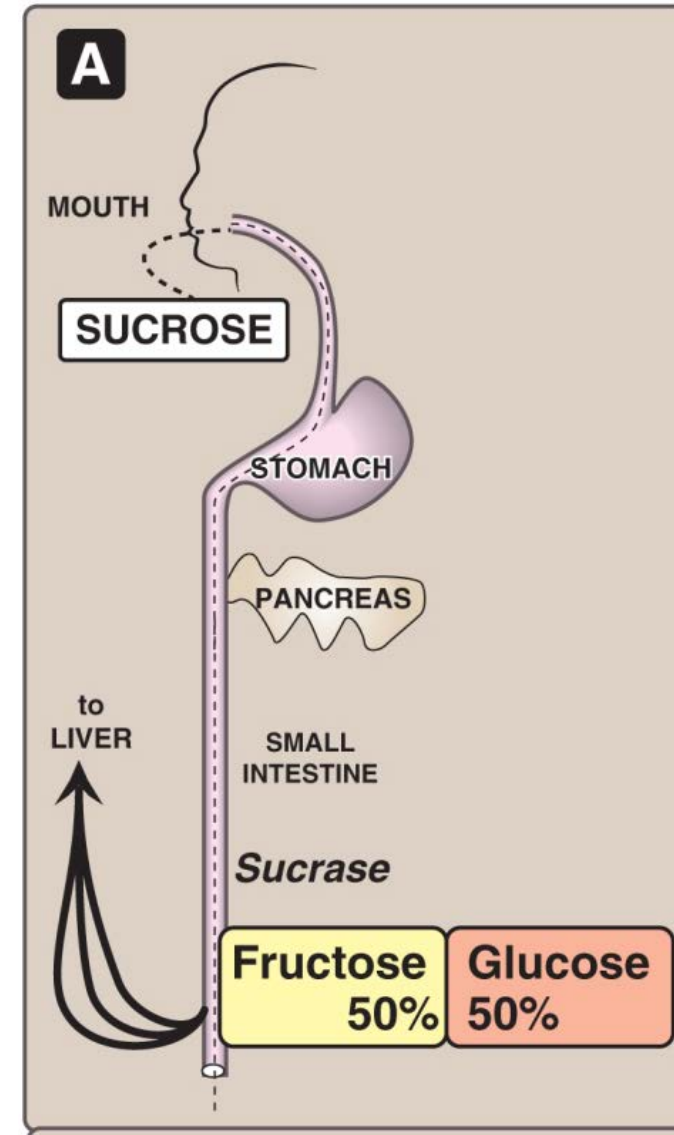


Lactose intolerance

Sucrose



- Glucose + Fructose
- Fructose : glucose = 1
- Non-reducing sugar: C1 and C2 not free
- Intestinal **sucrase** (brush border disaccharidase)
- Avoid: Hereditary fructose intolerance
- Refined sugars



High fructose corn syrup

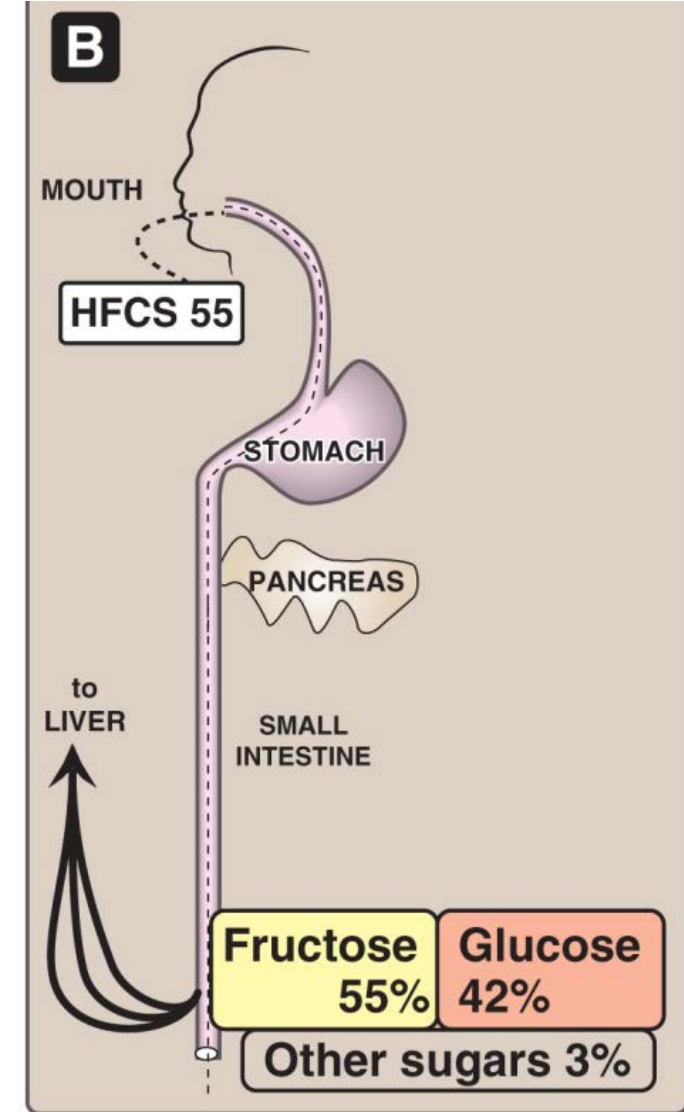
- Fructose (55%) and glucose (42%)
- **Fructose:** glucose ratio is greater than 1
- Mixture of monosaccharides
- Sweetener in soft drinks,

The Corny Truth About HIGH FRUCTOSE CORN SYRUP

Top 10 Foods with the Highest Quantity of HFCS:

- | | |
|---------------------|------------------------|
| 1) Yogurt | 6) Boxed Mac n Cheese |
| 2) Breads | 7) Salad Dressing |
| 3) Frozen Pizza | 8) Tomato-Based Sauces |
| 4) Cereal Bars | 9) Apple Sauce |
| 5) Cocktail Peanuts | 10) Canned Fruit |

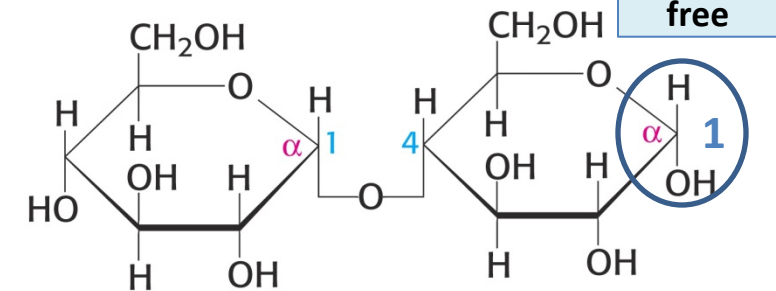
*High Fructose Corn Syrup has been linked directly to obesity, diabetes and metabolic dysfunction



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Maltose

- Two **glucose** units
- **$\alpha 1 \rightarrow 4$** glycosidic link
- Reducing sugar
- Intermediate product of starch digestion by amylase
- Intestinal **maltase** (brush border disaccharidase)



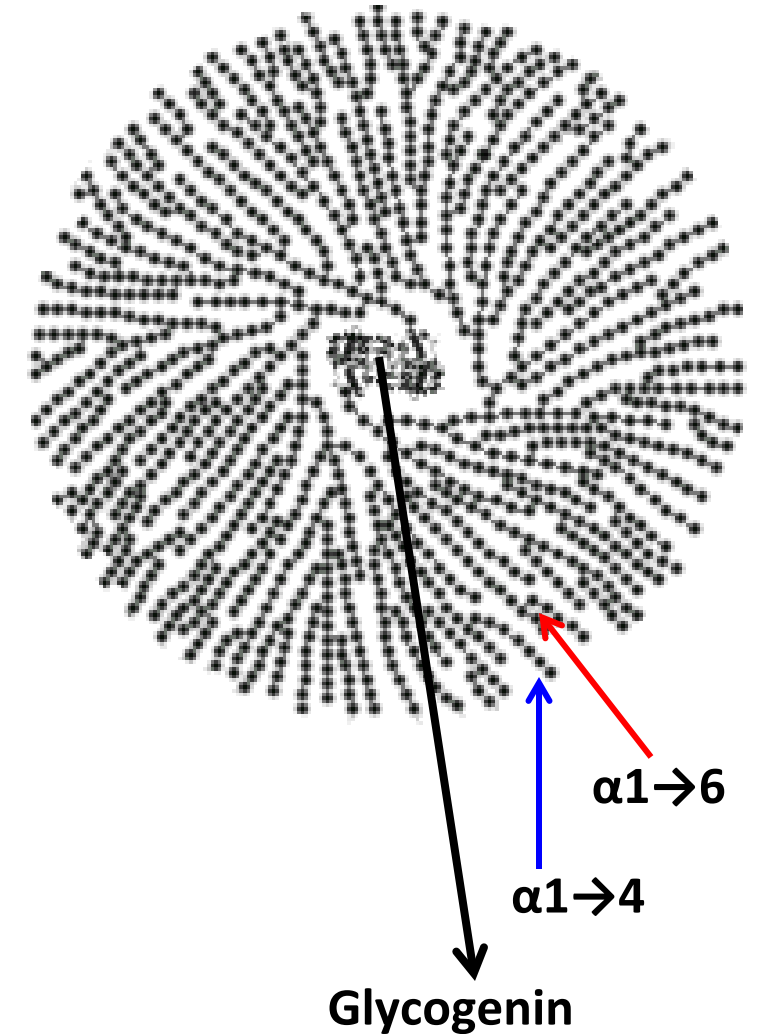
Maltose
(α -D-Glucopyranosyl-(1 $\rightarrow$ 4)- α -D-glucopyranose)

Polysaccharides

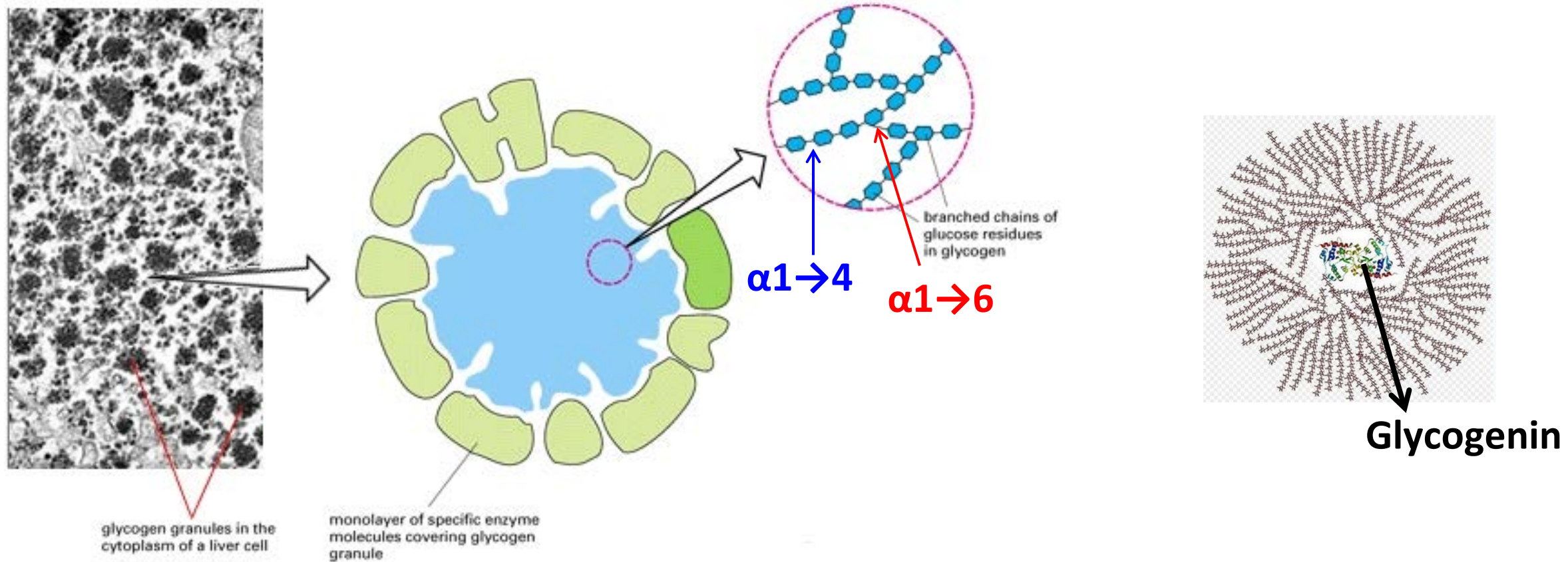
- **Homopolysaccharides**
 - Starch, glycogen and cellulose (glucose units)
- **Heteropolysaccharides**
 - Glycosaminoglycans/ Mucopolysaccharides

Glycogen (storage polysaccharide)

- Liver and muscle as glycogen granules
- Glucose units
- Linear chain $\alpha 1 \rightarrow 4$ glycosidic links
- Branch point $\alpha 1 \rightarrow 6$ glycosidic links
- Branch has 8-10 units
- **Glycogenin: Core protein**
- Glycogen storage disorders
 - Liver or muscle
 - Abnormal structure



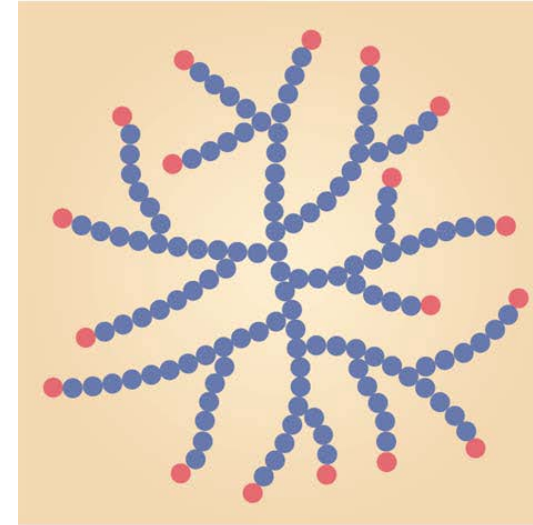
Glycogen granules in liver



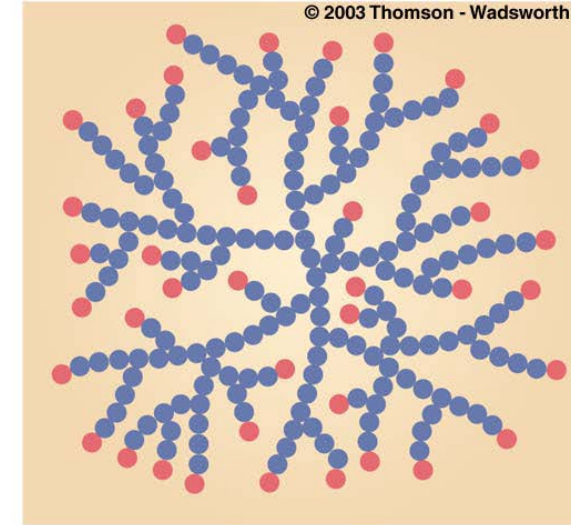
- Enzymes act on ends of branches
- Remove or add glucose

Starch (dietary polysaccharide)

- Two components:
 - Amylose (linear unbranched)
 - Amylopectin (branched)
 - Fewer longer branches
- Glucose units
- Linear chain $\alpha 1 \rightarrow 4$ glycosidic links
- Branch point: Amylopectin $\alpha 1 \rightarrow 6$ links
- Main dietary polysaccharide
- Salivary and pancreatic **amylase**



Amylopectin



Glycogen

Starchy Foods



Bread



Cereals



Pasta



Rice



Potatoes

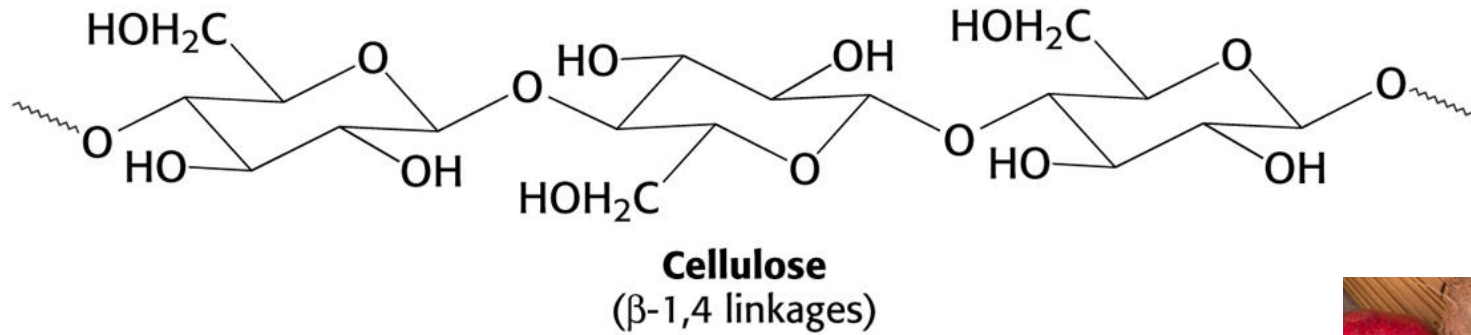


Beans

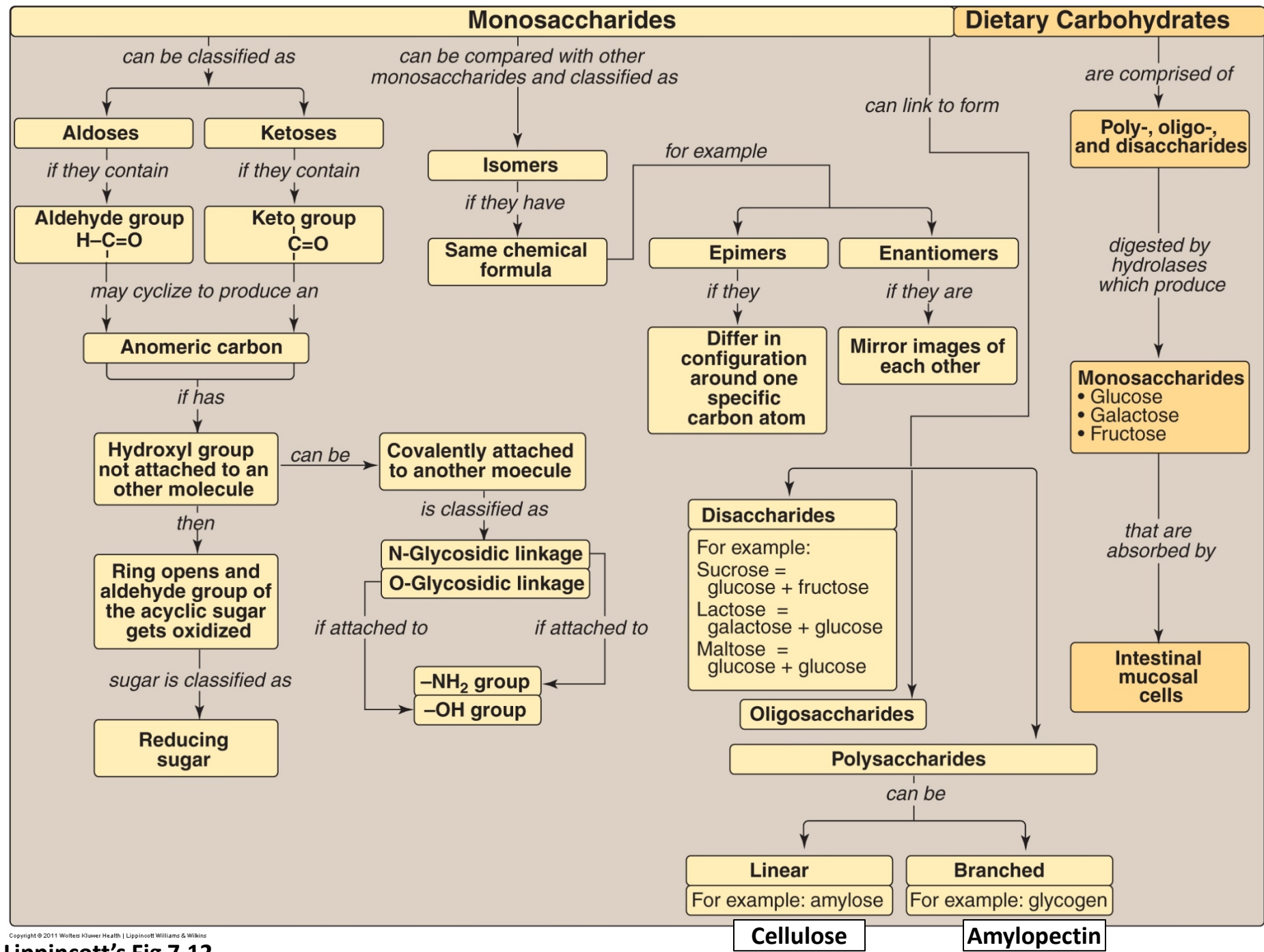


Chestnuts

Cellulose (dietary fiber)



- Undigestible carbohydrate
 - **Unbranched homopolysaccharide**
 - **Glucose** units **$\beta 1 \rightarrow 4$** links
 - Cannot be cleaved by digestive enzymes in the human gut
 - Lactase hydrolyzes $\beta 1 \rightarrow 4$ links only in lactose and NOT in cellulose!!
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Please email shupadhya@sgu.edu if you have any additional questions